

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to develop an intelligent **Flood Prediction System** using machine learning to improve the accuracy and timeliness of flood forecasting. The system addresses the limitations of traditional flood prediction methods by providing real-time predictions based on environmental parameters. Key features include a machine learning-based prediction model, an interactive web interface, and fast prediction results.

Project Overview	
Objective	To develop a machine learning-based flood prediction system that accurately predicts flood occurrence using environmental and weather-related parameters, enabling timely decision-making and disaster preparedness.
Scope	The project focuses on collecting flood-related data, training a machine learning model, and deploying it through a Flask web application for real-time flood prediction and user interaction.
Problem Statement	
Description	Existing flood prediction methods are often slow, less accurate, and unable to provide timely warnings. This affects disaster preparedness and increases the risk of property damage and loss of life.

Impact	An accurate flood prediction system helps improve disaster management, supports early warning systems, reduces risks to human life and property, and assists authorities in making informed decisions.
Proposed Solution	
Approach	Develop a supervised machine learning model trained on historical flood-related data and integrate it into a Flask-based web application to generate real-time flood predictions.
Key Features	- Machine learning-based flood prediction model. - Real-time prediction through a web interface. - User-friendly input form for environmental parameters. - Fast and accurate prediction results. - Easy deployment using the Flask framework.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor or T4 GPU (for model training)
Memory	RAM specifications	8 GB or above
Storage	Disk space for datasets, models, and application files	1 TB SSD

Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning & Data Processing Libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, joblib
Development Environment	IDE	Jupyter Notebook, Visual Studio Code or PyCharm
Data		
Data	Source, size, format	Kaggle Flood Prediction Dataset, CSV format (Historical flood and rainfall data)